

TESTING PLAN

EDUGENIE – AI-POWERED LEARNING ASSISTANT

TEST OBJECTIVES

1. Verify that the FastAPI application starts.
2. Verify that the frontend loads.
3. Verify that each learning module can be reached through its backend workflow.
4. Verify that AI-generated output is returned when the configured service/model is available.
5. Verify that the local explanation model can be loaded in an adequately provisioned environment.
6. Verify that credentials are not displayed in the application package.

FUNCTIONAL TEST AREAS

TC-01 Application Startup:

Expected: FastAPI starts successfully with the configured environment.

TC-02 Main Interface:

Expected: EduGenie web interface loads.

TC-03 Question Answering:

Input: Valid academic question.

Expected: Generated answer is returned.

TC-04 Concept Explanation:

Input: Valid academic concept.

Expected: Simplified explanation is returned.

TC-05 Quiz:

Input: Topic or learning material.

Expected: Multiple-choice quiz is generated.

TC-06 Summary:

Input: Long educational text.

Expected: Concise summary is generated.

TC-07 Learning Path:

Input: Learning topic.

Expected: Structured progression is generated.

TC-08 Invalid / Missing Input:

Expected: Application should handle the request without exposing credentials.

TEST ENVIRONMENT

- Python environment
- FastAPI/Uvicorn
- Web browser
- Gemini API connectivity for cloud functions
- Sufficient resources for the local language model